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Nc-miR403 positively regulates the iron deficiency stress response in *Poplar*

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ABSTRACT

Iron (Fe) deficiency is a widespread nutritional constraint that adversely affects plant growth and development. However, the roles of miRNAs in iron deficiency responses in fast-growing woody plants remain poorly understood. In this study, *Neolamarckia cadamba* was subjected to short-term iron deficiency treatment, and early root tissues were analyzed using whole-transcriptome sequencing to investigate its physiological and molecular responses. Among the identified miRNAs, Nc-miR403 was strongly induced under iron deficiency. To further explore its function, Nc-miR403 was overexpressed in 84K poplar (*Populus alba* × *P. tremula var. glandulosa*). Under iron-deficient conditions, Nc-miR403-overexpressing poplar intriguingly exhibited enhanced root growth, higher chlorophyll content, and increased net photosynthetic rate compared with wild-type plants. Additionally, the activities of superoxide dismutase (SOD) and peroxidase (POD) were significantly increased, indicating improved stress tolerance. Transcriptome analysis combined with weighted gene co-expression network analysis (WGCNA) identified several key modules associated with root growth and photosynthetic traits. These genes were mainly involved in mitochondrial iron transport, chlorophyll biosynthesis, and photosynthesis. Notably, iron homeostasis *(MIT2, NEET, and MSRB2*) and photosynthesis (*HEMA1, PORA, PSB27,* and *PSAF*) related genes were upregulated in Nc-miR403-overexpressing plants. Our findings firstly demonstrate that Nc-miR403 plays a positive regulatory role in iron deficiency responses and provide gene resources as well as valuable insights for improving iron deficiency tolerance in woody plants.

Introduction

Iron (Fe) is an essential micronutrient required for plant growth and development. As a cofactor of numerous redox proteins, iron participates in key metabolic processes including chlorophyll biosynthesis, photosynthesis and antioxidant defense system (Aksoy et al., 2013; Balk and Schaedler, 2014). However, in many soil, particularly in calcareous or alkaline soils, iron predominantly exists in the ferric [Fe(III)] form which is poorly available for plant uptake, thereby limiting plant growth and development (Guerinot and Yi, 1994).

Iron deficiency elicits a cascade of physiological and molecular responses in plants. The root system, as the primary organ for nutrient uptake, undergoes significant morphological and physiological changes under iron deficient conditions. These include increased root hair formation, root tip swelling, and enhanced lateral root development, all of which improve iron acquisition efficiency (Ning et al., 2023). In addition to morphological adaptations, iron deficiency severely affects photosynthesis. Chloroplast in mesophyll cells contains highest proportion of cellular iron, accounting for approximately 60%–80% of the total iron content (López-Millán et al., 2016). Consequently, impaired chlorophyll biosynthesis under iron deficiency leads to leaf chlorosis. Moreover, ultrastructural abnormalities in chloroplasts have been observed under iron-deficient conditions, including a reduction in the number of thylakoid membranes within the lamellar structure (Zuo and Zhang, 2009). Iron is an integral component of redox systems within photosystem I (PSI), photosystem II (PSII), cytochrome b6f complex, and ferredoxins (Zuo and Zhang, 2009; Zhang et al., 2025). Therefore, iron deficiency disrupts assembly of photosynthetic complexes and reduced electron transport efficiency, ultimately leading to a significant decline in net photosynthetic rate (Singh et al., 2005). Furthermore, iron plays a crucial role in respiration-related enzymes and antioxidant defense systems. Iron deficiency severely impairs the activities of key enzymes in

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the reactive oxygen species (ROS) scavenging system, including superoxide dismutase (SOD), peroxidase (POD), and ascorbate peroxidase (APX), thereby weakening the ability of plants to cope with against oxidative stress (Gupta et al., 2024).

To adapt to Fe-limited environment, plants have evolved two main strategies for iron acquisition from the rhizosphere: Strategy I, a reduction-based mechanism used in both dicots and non-graminaceous monocots; and Strategy II, a chelation-based mechanism used in graminaceous monocots (Kobayashi and Nishizawa, 2012). In *Arabidopsis thaliana*, Strategy I involves the reduction of Fe(III) to Fe(II) by the plasma membrane-localized *ferric chelate reductase FRO2*, followed by uptake of *Fe(II)* into root cells via *IRON REGULATED TRANSPORTER1 (IRT1)* (Robinson et al., 1999; Satbhai et al., 2017; Vert et al., 2002). Long-distance iron transport in Arabidopsis is mediated by several key components, including *FERRIC REDUCTASE DEFECTIVE 3 (FRD3), YELLOW STRIPE-LIKE (YSL)* transporters such as *YSL1* and *YSL3* (Chu et al., 2010), and *OLIGOPEPTIDE* transporter 3 (*OPT3*) (Mendoza-Cózatl et al., 2014).

The transcription factors (TFs) also play important roles in regulating plant responses to Fe deficiency, particularly members of the basic helix–loop–helix (bHLH) family members, which are involved in Fe uptake, transport, storage and other processes (Zhang et al., 2023a). In *Arabidopsis*, the *FER-like iron deficiency-induced transcription factor AtFIT1 (AtbHLH29)* forms heterodimers with *bHLH* proteins (*AtbHLH38, AtbHLH39, AtbHLH100*, and *AtbHLH101*), thereby activating the expression of *AtFRO2* and *AtIRT1*, which are essential for Fe acquisition (Yuan et al., 2008; Wang et al., 2013). *BTS* is considered a potential iron-sensing protein in *Arabidopsis thaliana*, which can bind to iron ions and ubiquitinate to degrade *bHLH* IVc transcription factors (*bHLH105* and *bHLH115*) (Selote et al., 2015). The transcription factor *POPEYE* (*PYE*) and its homologues (*bHLH34, bHLH104, bHLH105/ILR3*, and *bHLH115*) further regulate Fe homeostasis through homo- or heterodimer formation (Ivanov et al., 2012) (Li et al., 2016). Under Fe-deficient conditions, upregulation of the *AtPYE* promotes root growth and inhibits the expression of several metal homeostasis genes, including *AtFRO3, AtNAS4* and *AtZIF1* (Long et al., 2010).

Beyond transcriptional regulation, plant hormones and signaling molecules also contribute iron deficiency-induced morphological and physiological responses (Ivanov et al., 2012). In parallel, iron deficiency also induces differential expression of non-coding RNAs, including microRNAs (miRNAs), long non-coding RNAs (lncRNAs), and circular RNAs (circRNAs) (Waters et al., 2012). MiRNAs are key post-transcriptional regulators that modulate gene expression by directing sequence-specific mRNA cleavage or translational repression of their target genes (Cui et al., 2020). Increasing evidence suggests that miRNAs fine-tune plant responses to iron deficiency by regulating transcription factors and downstream target genes (Kaya, 2025). For example, in tomato (*Solanum lycopersicum*), the SlymiR157–SlSPL-CNR module represents a miRNA-mediated regulatory pathway involved in Fe deficiency responses. (Zhu et al., 2022). In addition, miR408 regulates laccase-like multicopper oxidase genes (*LAC3, LAC12,* and *LAC13*), affecting vascular lignification and ferroxidase activity under Fe-limited conditions (Carrió-Seguí et al., 2019). Similarly, miR164 targets the *NAC5–NFYA8* module, thereby influencing root architecture, ferric reductase activity, and the expression of iron uptake genes such as *IRT1* and *FRO2* (Du et al., 2022) . An increasing number of miRNAs involved in Fe homeostasis have been identified across plant species. In *Arabidopsis thaliana*, eight microRNAs including miR397a, miR398a, and miR398b/c were downregulated under Fe deficiency and regulate Cu-containing proteins, suggesting their roles in adaptation to Fe limitation (Waters et al., 2012). In *citrus*, Fe deficiency alters the expression of miRNAs including miR172, miR395, miR319, miR398, miR397, and miR408, thereby affecting stress tolerance, sulfur and copper homeostasis, and lignin biosynthesis (Jin et al., 2021). In hexaploid wheat, high-throughput siRNA profiling identified more than 100 Fe-responsive miRNAs, including novel species exhibiting root- or shoot-specific expression (Sharma et al., 2024).

Globally, over one-third of terrestrial soils are iron (Fe)-deficient. Severe iron deficiency occurs widely across the North American continent, the Mediterranean coastal zones and parts of South America, and such soil iron insufficiency is also prevalent throughout China (Ning et al., 2023). Fe-deficient soils span extensive geographic ranges in China, stretching from the Sichuan Basin in the south to the Inner Mongolia Plateau in the north, and from the North China Plain in the east to the Loess Plateau in the west (Zou et al., 2001). Collectively, Fe deficiency predominates in arid and semi-arid regions of northern China. Soils in these areas are mostly calcareous and alkaline, characterized by elevated pH and bicarbonate concentrations, which severely constrain iron acquisition in Strategy I plant species and readily trigger iron-deficiency chlorosis in crops (Chen et al., 2019a; Zou et al., 2008; He, 2002) . Poplar, a typical Strategy I plant species, relies on rhizosphere acidification and ferric-chelate reduction for iron acquisition and is therefore particularly susceptible to Fe deficiency under alkaline soil conditions (Masuda et al., 2018). Alongside the continuous expansion of poplar plantations and the widespread application of novel poplar genotypes, Fe deficiency-triggered leaf chlorosis has become progressively aggravated in poplar. As a non-parasitic physiological disorder, poplar Fe chlorosis is primarily attributed to the low bioavailability of iron in forest soils (Wang, 2007).

In this study, *Neolamarckia cadamba* were subjected to short-term iron-deficiency treatment, and early root tissues were analyzed using whole-transcriptome sequencing, including mRNA, miRNA, lncRNA, and circRNA. Transcriptome analysis identified three iron deficiency–responsive miRNAs: miR403, miR-N153, and miR-N263, of which miR403 showed the most significant and sustained response to iron deficiency in *N.cadamba*. To investigate role of Nc-miR403 in regulating Fe homeostasis in woody plants, miR403 was overexpressed in 84K poplar (*Populus alba* × *Populus glandulosa*) using established transformation techniques (Tuskan et al., 2006). Our results firstly showed that Nc-miR403 positively regulates the response to iron deficiency comparing with wild-type plants. These findings provide new insights into miRNA-mediated regulatory mechanisms underlying iron deficiency responses in woody species, and offer a promising strategy for targeted breeding of poplar varieties with superior iron deficiency stress tolerance.

Materials and methods

Plant materials and iron deficiency treatment

All experiments were conducted in a cultivation room at 25°C with 16 h light and 8 h dark photoperiod at College of Forestry and Landscape Architecture, South China Agricultural University. Superior clones of *N. cadamba* were selected through large-scale screening and selected plants were cultivated in 1/4 Hoagland nutrient solution (Liu et al., 2021; Dai et al., 2020). Uniform seedlings were selected and transplanted into 2.5 L hydroponic boxes (39 cm length × 12 cm width × 10 cm height). Each box was equipped with 1 cm^3sponge blocks and foam boards to fix seedlings, with five seedlings per box serving as one biological replicate. Three independent replicates were established for both the control and treatment groups. For iron deficiency treatment, NaFe-EDTA was omitted from the solution and pH was adjusted to 5.8. The iron concentration in iron deficient treatment (FD) was 0 mM while the complete nutrient solution containing iron served as the control (CK). The roots samples were collected at the 12 h of treatment for whole transcriptome sequencing, while leaves, roots and stems were harvested after 14 and 21 weeks of treatment for physiological analysis. All experiments were repeated at least three times.

Tissue cultured seedlings of transgenic 84K poplar lines with uniform growth were selected, with wild-type plants were used as controls. The culture medium attached to the roots was removed by washing with tap water followed by rinsing with deionized water. After three weeks of

sand culture, the seedlings were treated with nutrient solutions containing different iron concentrations. Iron-free MS medium was used as the basal culture solution, and serial concentrations of NaFe-EDTA were added exogenously to generate nutrient solutions with different iron levels. Preliminary cultivation and phenotyping revealed distinct growth phenotypes of 84K poplar under media with divergent iron supplies. Accordingly, three iron regimes were designed: normal iron control (CK, 2/5-strength MS iron level, 14.68 mg·L^{-1}), moderate iron deficiency stress (1/2Fe, 7.34 mg·L^{-1}), and complete iron deprivation stress (-Fe, 0 mg·L^{-1}). The growth substrate consisted of iron-depleted pure quartz sand, sterilized at 121°C for 70 min. Plants were grown in an artificial greenhouse under light intensity of 3500 lx with a photoperiod of 16 h. The temperature was maintained at 25°C, and the relative humidity was kept at 45–55%.

Measurement of physiological parameters

Samples were randomly collected at 14 and 21 days under both control and iron deficient conditions. Plant height, fresh weight and dry weight were measured, and phenotypic changes were recorded and documented through photographs. Root architecture was analyzed using root scanner (Epson Expression 12000XL) and parameters including root length, root volume and root surface area was calculated using Win RHIZO software.

For dry weight determination, seedlings were rinsed three times with deionized water, blotted dry, and weighed to obtain fresh weight. Samples were then oven-dried to constant weight and reweighed. Each measurement was performed with three biological replicates.

The activities of enzyme, including Superoxide dismutase (SOD), Peroxidase (POD) and Ascorbate oxidase (APX) from leaves and roots were determined using commercial assay kits (Suzhou Kemin Biotechnology Co (Suzhou, China)) following manufacturer's instructions (Yao et al., 2017).

Chlorophyll content and fluorescence determination

Chlorophyll fluorescence was measured in vivo using the IMAGING-PAM chlorophyll fluorescence imaging system (WALZ, Germany) and fluorescence parameters were analyzed accordingly. Chlorophyll content was determined spectrophotometrically using 80% acetone extraction method. Approximately 0.1 g of leaf tissue (second fully expanded leaf) was collected, veins were removed and weighed in a 10ml centrifuge tube. Five milliliters of 80% acetone was added, and samples were incubated in the dark for one week until complete pigment extraction. Chlorophyll content was measured in triplicate via spectrophotometry, following the protocol described by(Wellburn, 1994).

Element content determination

Root samples were ground into powder and oven-dried at 80°C to constant weight. Approximately 0.5 g of dried sample was transferred into digestion tubes, and 5 mL HNO_3 was added for overnight predigestion. Subsequently, 2 mL of 30% H_2O_2 was added, and the samples were digested using a microwave digestion system (ramp program; final temperature ≥180°C for 20 min).

After cooling, the digested solutions were transferred to 50 mL volumetric flasks, diluted to volume with deionized water, and filtered through quantitative filter paper. The concentrations of Fe, Cu, Mn, and Zn were determined using an inductively coupled plasma atomic emission spectrometer (ICP-AES; VARIAN 710-ES).

Photosynthetic measurements

Photosynthetic parameters of poplar plants grown under iron-deficiency for two months were measured using a LI-6400XT portable photosynthesis system (Li-Cor, USA). Measurements were conducted at 25°C under a photosynthetically active radiation (PAR) of 2000 μmol m^{-2} s^{-1}. Healthy, fully expanded leaves with similar growth status were selected and data from three leaves from the same node of each plant were selected for measurement, and four readings were recorded per leaf.

RNA extraction, library preparation, and sequencing

Root samples from plants treated under FD and CK conditions for 12 h were collected, with three biological replicates per treatment. Samples were frozen in liquid nitrogen and ground into powder prior to RNA extraction. RNA quality, purity, and integrity were assessed using 1% agarose gel electrophoresis, a NanoPhotometer® spectrophotometer (IMPLEN, USA), and an Agilent 2100 Bioanalyzer with the RNA Nano 6000 kit (Agilent Technologies, USA).

Strand-specific libraries for mRNA, lncRNA, and circRNA were constructed using ribosomal RNA removal methods, while miRNA libraries were prepared using the Small RNA Sample Prep Kit. Whole-transcriptome sequencing and and miRNA-seq of *N. cadamba* were both performed by Novogene (Beijing, China), the whole-transcriptome data are available in GSA(CNCB-NGDC) with accession CRA041569, and miRNA-seq data have been deposited in GSA (CNCB-NGDC) under temporary submission ID subCRA068163, and will be publicly accessible after paper publication.

Poplar RNA-seq was conducted by Biomarker (Qingdao, China) using the Illumina HiSeq 2000 platform, the raw transcriptome sequencing data have been deposited in GSA (CNCB-NGDC) under temporary submission ID subCRA068852, and will be publicly accessible after paper publication.

Raw reads were processed to remove adapters, poly-N sequences, and low-quality reads. Clean reads were assembled using Trinity software to generate a non-redundant unigene dataset. Functional annotation was performed by aligning sequences against the Pfam database using HMMER.

Differential gene expression and functional analysis

Gene expression levels were calculated as fragments per kilobase of transcript per million mapped reads (FPKM). Genes with a false discovery rate (FDR) < 0.01 and $|\log_2(\text{fold change})| \geq 2$ were considered differentially expressed genes (DEGs).

Log_2-transformed FPKM values were used for K-means clustering based on Pearson correlation using Microarray Experiment Viewer (MeV, v4.3). Heatmaps were generated using TBtools. Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) enrichment analyses were conducted using the ClusterProfiler package in R.

Weighted gene co-expression network analysis (WGCNA) was performed using the WGCNA package in R, and statistical significance was determined based on correlation p-values.

Quantitative qRT-PCR analysis

cDNA synthesis was performed using the TaKaRa PrimeScript™ RT Reagent Kit with gDNA Eraser (Takara, Japan) for mRNA and the TransScript® miRNA First-Strand cDNA Synthesis SuperMix (TransGen Biotech) for miRNA. Quantitative PCR was conducted using a LightCycler 480 system. In *N.cadamba*, SAMDC was selected as the internal reference gene for mRNA, and U6 served as the internal control for miRNA. In poplar, Actin was used as the reference gene for mRNA quantification. RT-qPCR data were analyzed using the comparative Ct ($2^{-\Delta\Delta Ct}$) method. The Ct value of each target gene was normalized against the housekeeping gene to obtain ΔCt (ΔCt = Ct_target − Ct_reference). Subsequently, ΔΔCt values were calculated as ΔCt_sample − ΔCt_control, and relative expression levels were determined as $2^{-\Delta\Delta Ct}$. Each sample included three biological replicates, and each assay was performed with at least three technical replicates. Primer sequences are

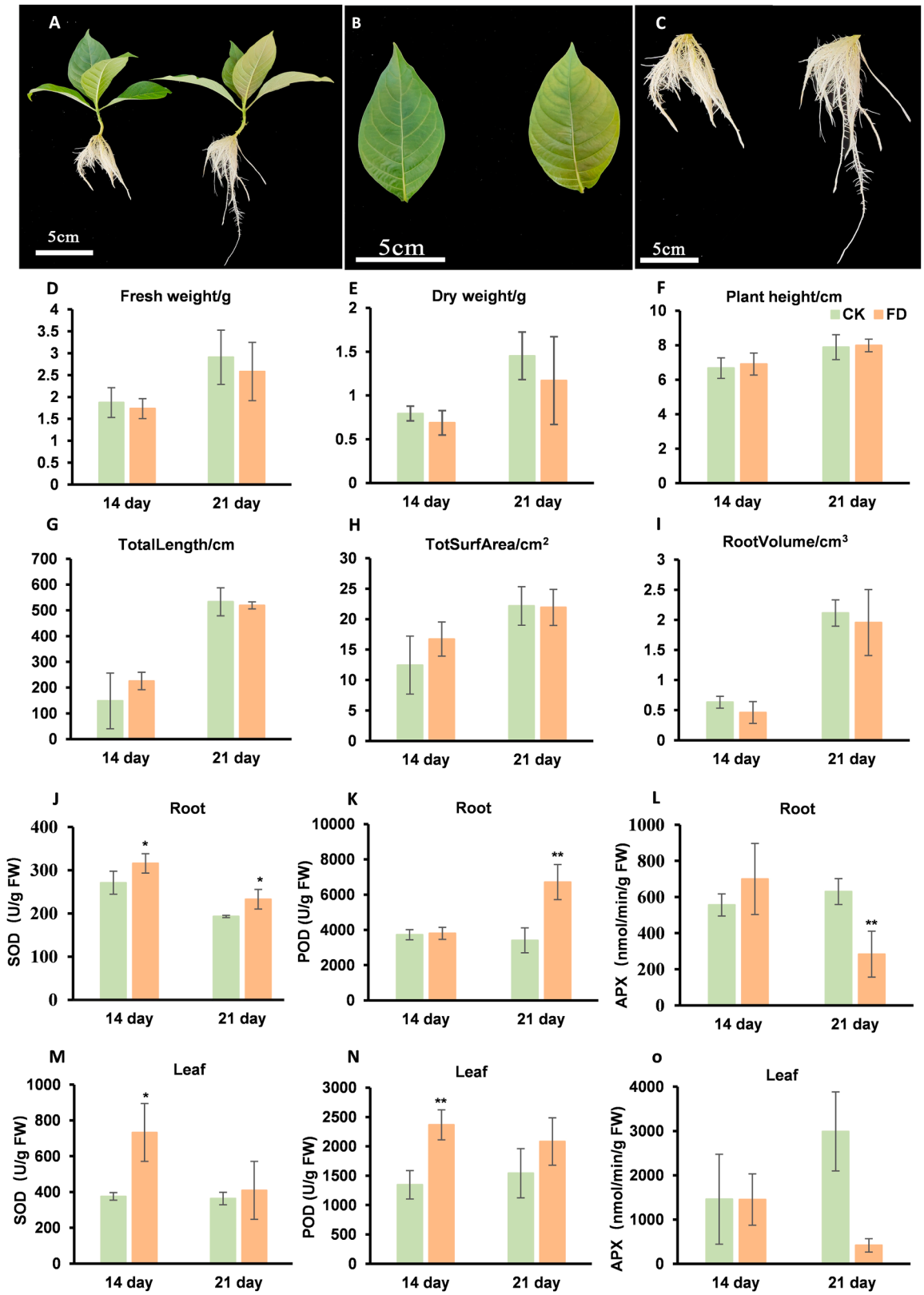


Fig. 1. Iron deficiency inhibits the growth and development of *N. cadamba*. (A–C) Phenotypic changes in the whole plant, roots, and leaves after 9 days of iron-deficiency treatment. (D–F) Fresh weight, dry weight, and plant height of control (CK) and Fe-deficient (FD) plants at 14 d and 21d. (G–I) Total root length, total root surface area, and total root volume of CK and FD plants at 14d and 21d. (J–L) Activities of superoxide dismutase (SOD), peroxidase (POD), and ascorbate peroxidase (APX) in roots of CK and FD plants at14d and 21d. (M–O) Activities of SOD, POD, and APX in leaves of CK and FD plants at 14d and 21d. Data are presented as the mean ± SD (n = 3). Differences between the mean values of CK and FD were compared using the T-test (* indicates a significant difference at $p < 0.05$; ** indicates a significant difference at $P < 0.01$).

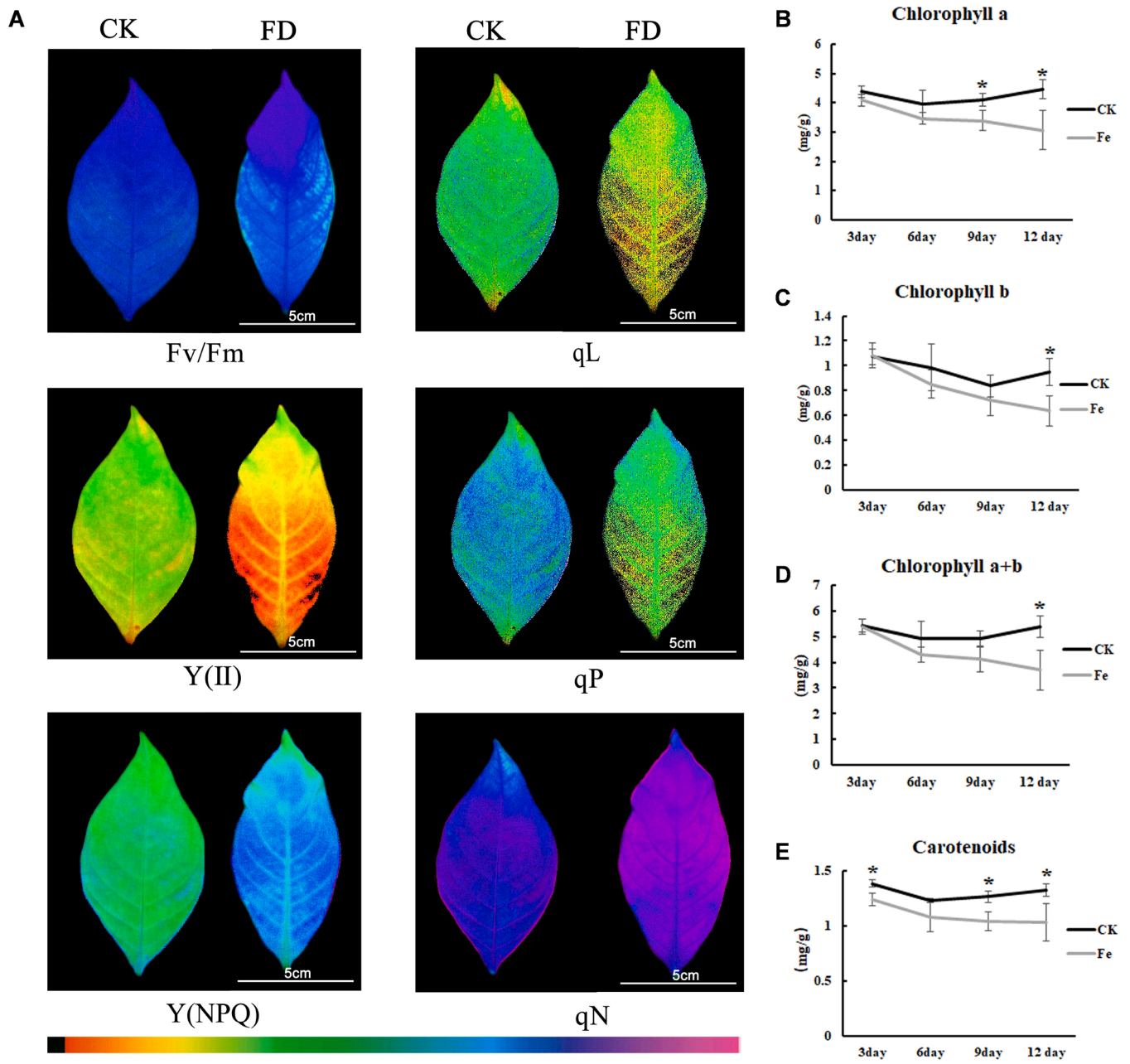


Fig. 2. Iron deficiency reduces photosynthesis in *N. cadamba*. (A) Chlorophyll fluorescence imaging under control (CK) and Fe-deficient (FD) conditions. (B–E) Changes in chlorophyll a, chlorophyll b, total chlorophyll, and carotenoid contents under CK and FD conditions. Data are presented as the mean ± SD (n = 3). Differences between the mean values of CK and FD were compared using the *t*-test (* indicates a significant difference at $p < 0.05$).

listed in Table S1.

Generation of transgenic poplar and overexpression (OE) analysis

To investigate the function of Nc-miR403, its precursor sequence was cloned into the pK2GW7 expression vector. The recombinant construct was introduced into *Agrobacterium tumefaciens* strain GV3101 and used to transform poplar via the leaf-disc method. Transgenic lines were confirmed by PCR and RT–qPCR using primers listed in Table S1. Two independent overexpression lines with the highest expression levels (#5 and #7) were selected for further phenotypic and physiological analyses.

Result

Morphology and growth parameters of N. cadamba under Fe deficiency

After 9-d of Fe-deficiency treatment, *N. cadamba* exhibited typical symptoms of iron deficiency, including interveinal chlorosis, elongated primary roots, and increased lateral root formation compared with control plants (Fig. 1A–C). Further growth analysis at 14d and 21d revealed reductions in fresh and dry weight whereas plant height remained largely unaffected (Fig. 1D–F). In contrast, Fe deficiency markedly altered root system architecture. At 14d, total root length and root surface area increased by 52.27% and 34.28%, respectively; however, both parameters exhibited a slight decline at 21d. These results suggest that Fe deficiency initially stimulates root development but ultimately constrains plant growth under prolonged stress (Fig. 1G–I).

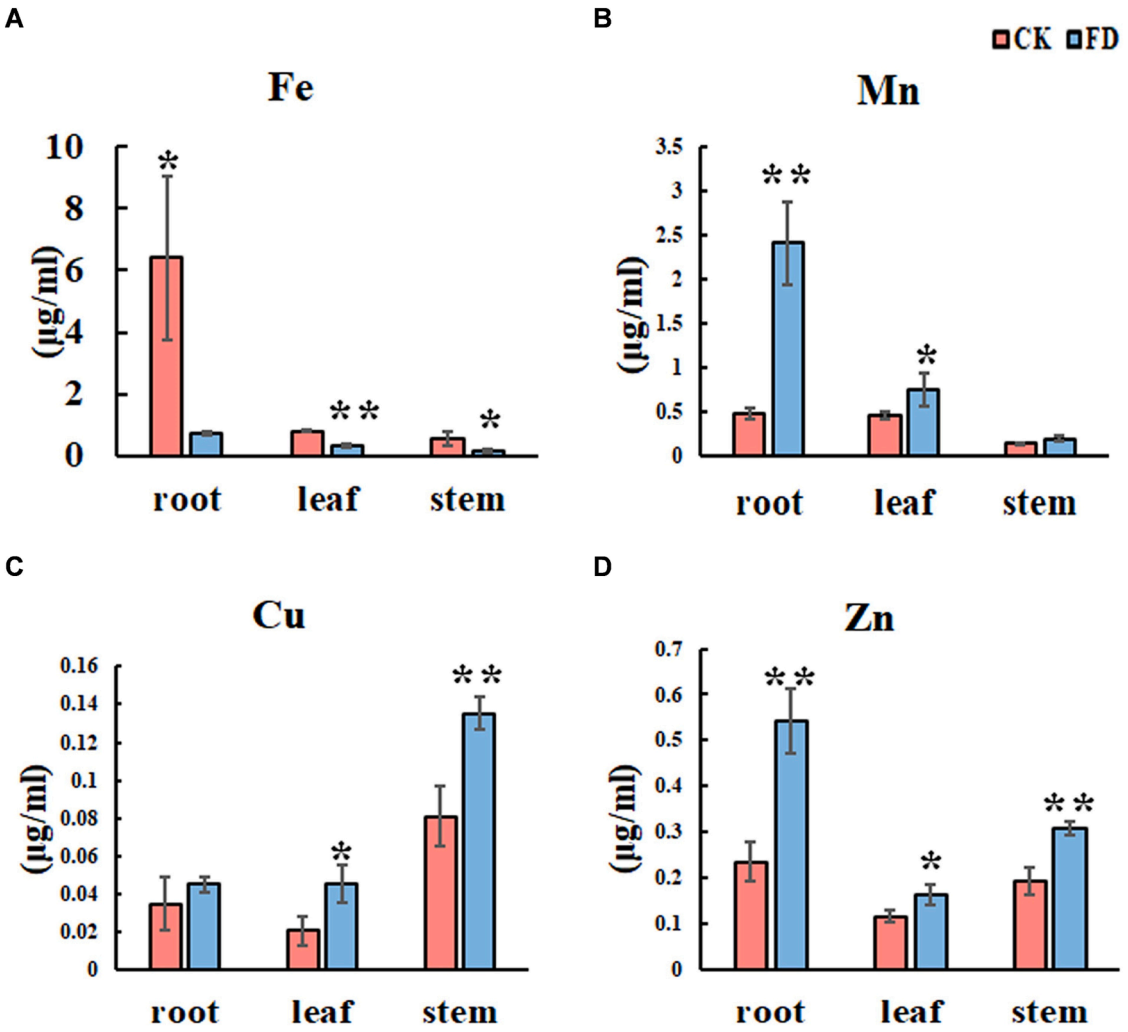


Fig. 3. Effects of iron deficiency on metal element contents in *N. cadamba*. (A–D) Measurement of Fe, Mn, Cu, and Zn contents in N. cadamba after 3 weeks of iron-deficiency treatment. Data are presented as the mean ± SD (n = 3). Differences between the mean values of CK and FD were compared using the t-test (* indicates a significant difference at $p < 0.05$; ** indicates a significant difference at $P < 0.01$).

Fe deficiency also significantly influenced antioxidant enzyme activities. Compared with CK plants, SOD and POD activities were significantly increased, whereas APX activity significantly decreased. At 21d, root SOD and POD activities significantly increased about 20.64% and 97.06%, respectively while APX activity decreased by about 54.99% (Fig. 1J–O). Collectively, these results indicate that Fe deficiency significantly alters root morphology and antioxidant responses, ultimately affecting the growth and development of *N. cadamba*.

Effects of iron deficiency on photosynthesis in N. cadamba

Newly emerged leaves began to show chlorosis at 9d under Fe-deficient conditions in *N. cadamba*. Chlorophyll fluorescence imaging using an Imaging-PAM system revealed that the maximum quantum yield of photosystem II (PSII) (Fv/Fm), light-harvesting antenna (qL), the effective photochemical quantum yield of PSI [Y(II)], and photochemical quenching coefficient (qP) were decreased significantly. In contrast, non-photochemical fluorescence quenching [Y(NPQ)] and non-photochemical quenching of chlorophyll fluorescence (qN) increased markedly, indicating that iron deficiency inhibited electron transport in PSII and reduced photosynthetic efficiency (Fig. 2A). Consistent with these observations, leaf pigment analysis showed that chlorophyll a, chlorophyll b, carotenoids, and total chlorophyll contents were significantly reduced at 12d of treatment (Fig. 2B–E), suggesting that iron deficiency disrupts chloroplast function and weakens light-harvesting capacity, ultimately suppressing photosynthesis in *N. cadamba*.

Effects of Fe deficiency on the accumulation of other metal elements in N. cadamba

After 21d of Fe-deficiency treatment, significant changes in mineral elements concentrations were observed (Fig. 3A-D). As expected, Fe content was decreased markedly in all examined tissues, with reduction of 88.66%, 73.57%, and 57.12% in roots, stems, and leaves, respectively. In contrast, the concentrations of Mn, Cu, and Zn were increased under Fe-deficient conditions. Specifically, Mn content increased significantly in roots and leaves by 415.50% and 68.00%, respectively while Cu content increased significantly by 121.17% in leaves and 67.30%.in stems. These results indicate that Fe deficiency not only leads to a substantial reduction in Fe accumulation but also promotes the uptake and accumulation of other divalent metal ions, suggesting a reprogramming of metal homeostasis under Fe-deficiency stress.

Whole-transcriptome analysis to identify differentially expressed RNAs under iron deficiency

To investigate the early response of *N. cadamba* to iron deficiency,

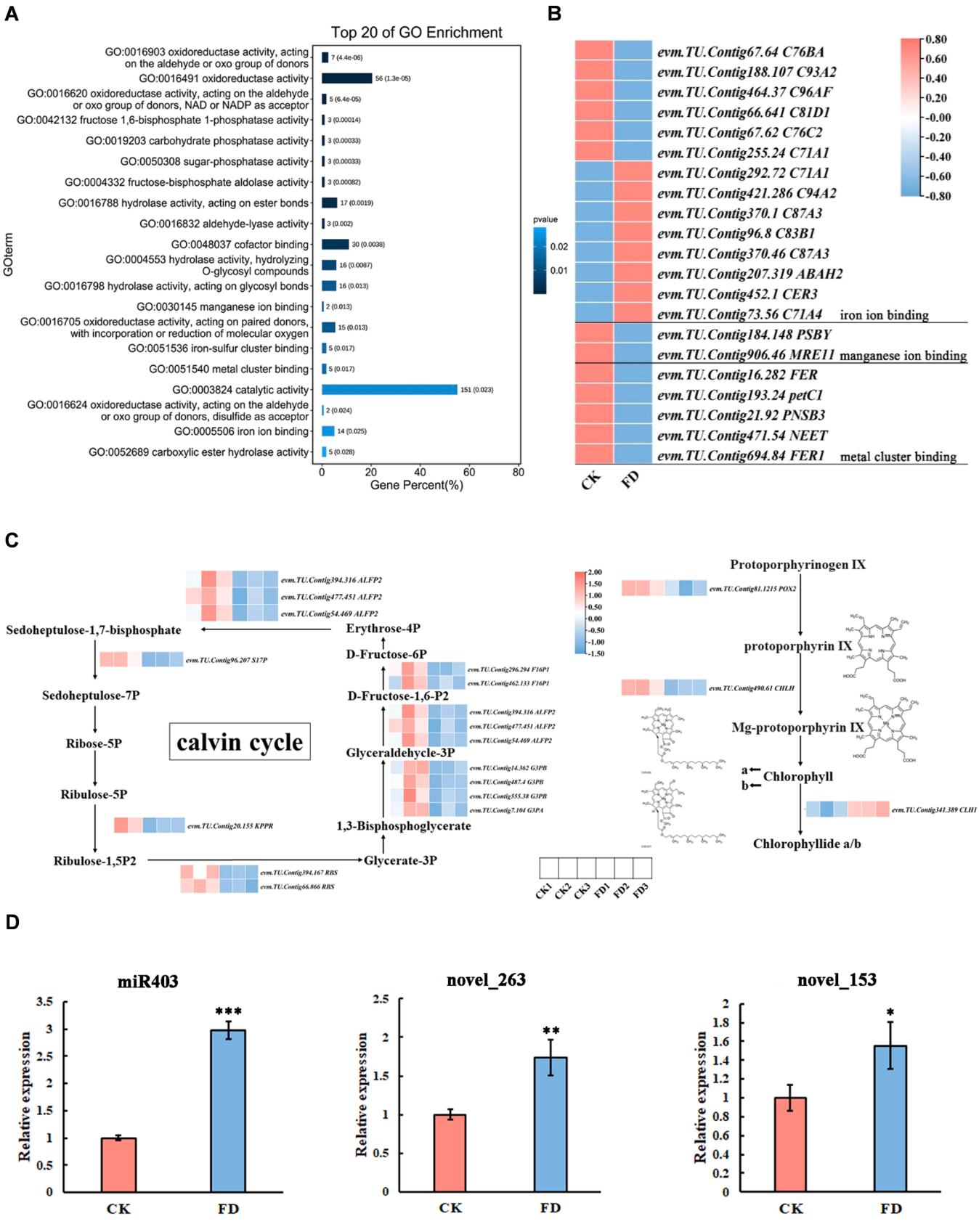


Fig. 4. Transcriptomic responses of *N. cadamba* to iron deficiency. (A) GO enrichment analysis of differentially expressed genes (DEGs) under control (CK) and Fe-deficient (FD) conditions. (B) Heatmap showing the expression patterns of iron deficiency-responsive genes under CK and FD conditions. (C) Changes in gene expression involved in the Calvin cycle and chlorophyll biosynthesis pathways under CK and FD conditions. (D) qRT–PCR validation of the relative expression levels of miR403, novel_153, and novel_263 under CK and FD conditions.

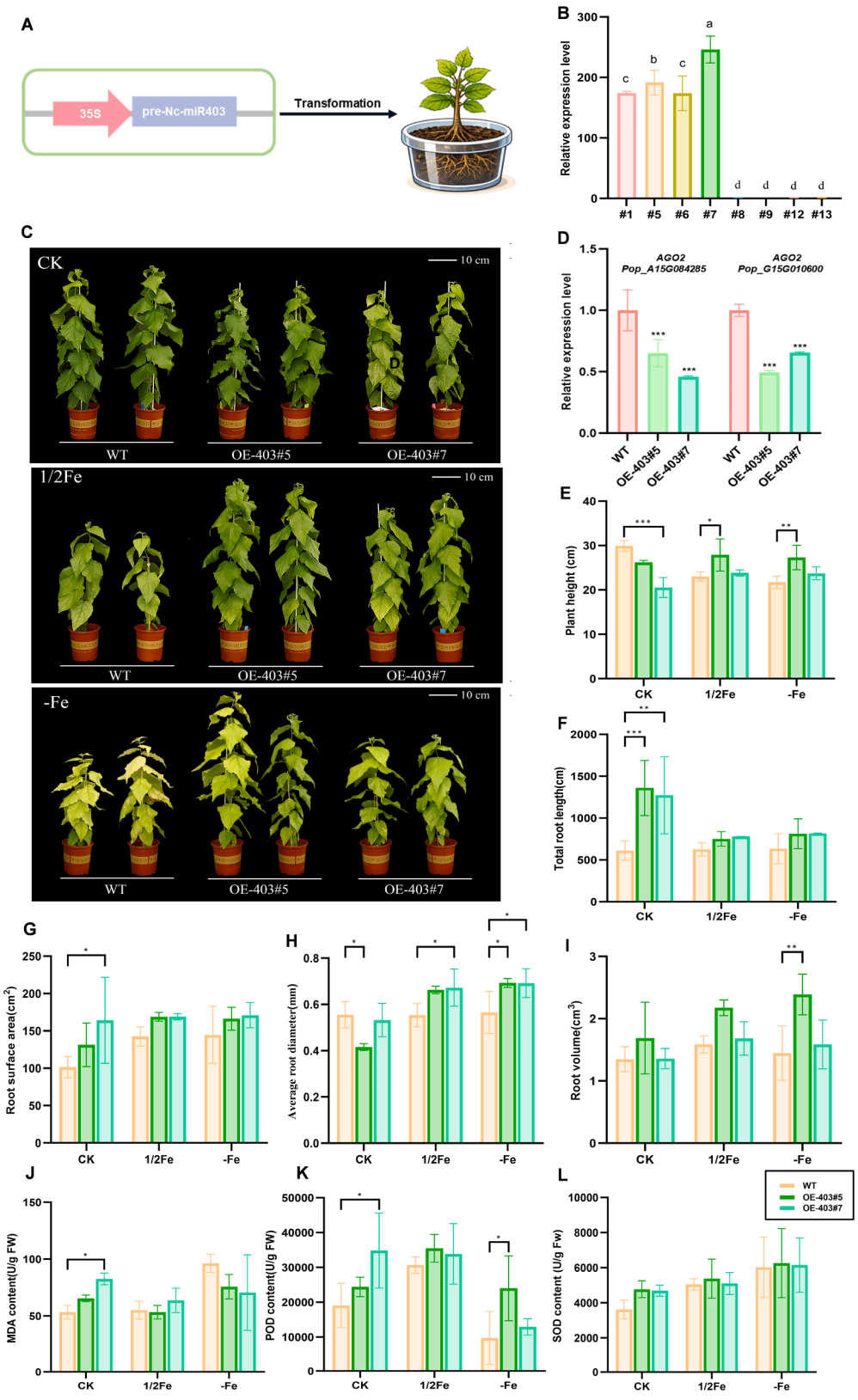


Fig. 5. Overexpression of Nc-miR403 enhances iron-deficiency tolerance in poplar. (A) Schematic of the Nc-miR403 precursor overexpression construct and poplar transformation. (B) Relative expression levels of miR403 in OE-miR403 lines. (C) Phenotypes of OE-miR403 and WT plants under CK, 1/2 Fe, and −Fe treatments. (D) Relative expression levels of *AGO2* in OE-miR403 and WT poplar plants. (E) Plant height of OE-miR403 and WT under different Fe concentrations. (F–I) Total root length (cm), total root surface area (cm^2), average root diameter (cm), and total root volume (cm^3) of OE-miR403 and WT under different Fe concentrations. (J–L) MDA content, POD activity, and SOD activity of OE-miR403 and WT under different Fe concentrations. Data are presented as means ± SD of three independent biological replicates. Different letters indicate significant differences among treatments (* indicates a significant difference compared with WT at $P < 0.05$; ** indicates a significant difference at $P < 0.01$; *** indicates a significant difference at $P < 0.001$).

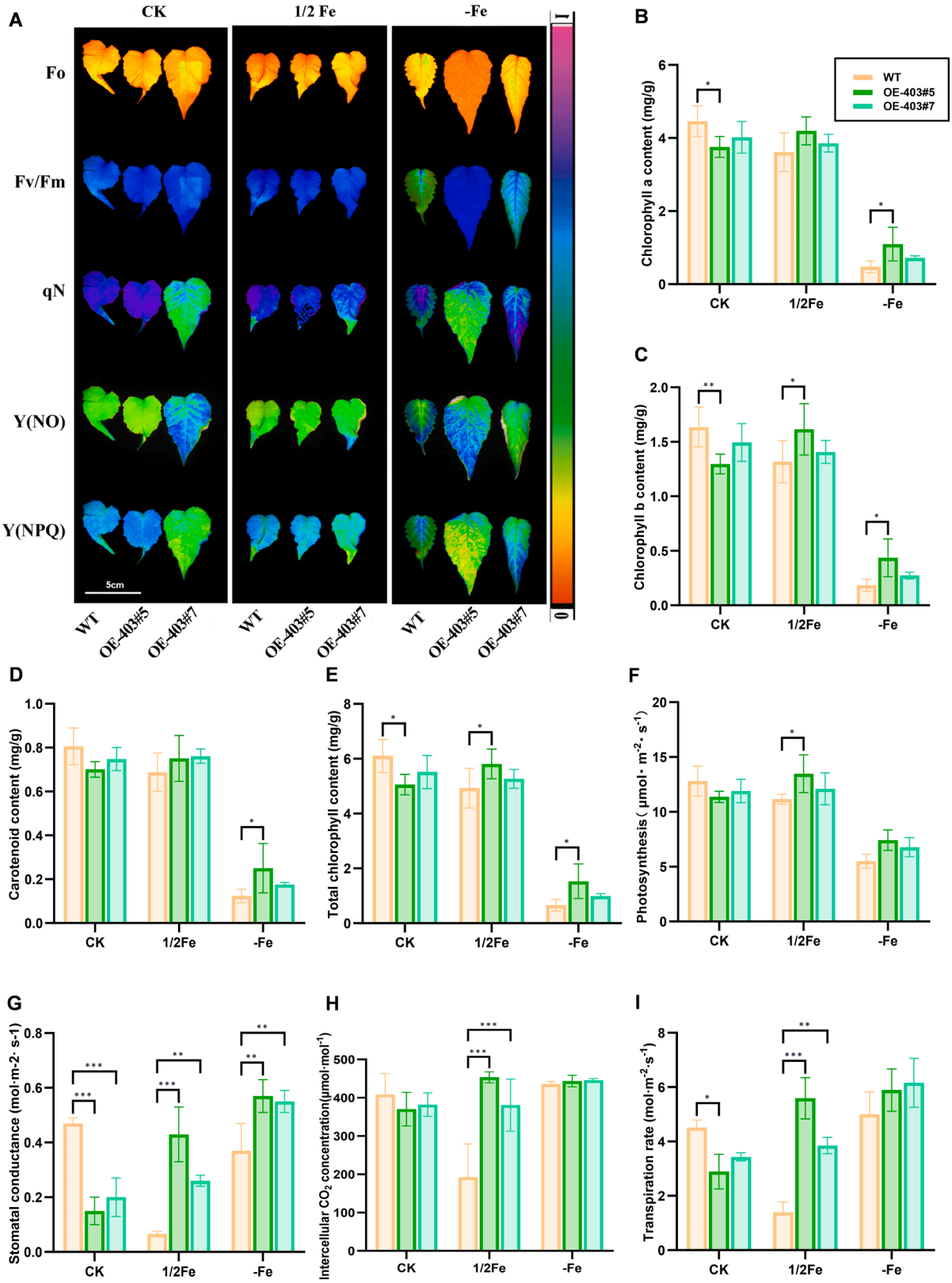


Fig. 6. Overexpression of Nc-miR403 enhances photosynthesis in poplar under iron-deficiency stress. (A) Chlorophyll fluorescence imaging of OE-miR403 lines and wild-type (WT) plants. (B–E) Chlorophyll a content, chlorophyll b content, carotenoid content, and total chlorophyll content of OE-miR403 and WT under different Fe concentrations. (F–I) Net photosynthetic rate (Pn), stomatal conductance (Gs), intercellular CO_2 concentration (Ci), and transpiration rate (Tr) of OE-miR403 and WT under different Fe concentrations. Data are presented as means ± SD of three independent biological replicates (* indicates a significant difference compared with WT at $P < 0.05$; ** indicates a significant difference at $P < 0.01$; *** indicates a significant difference at $P < 0.001$.).

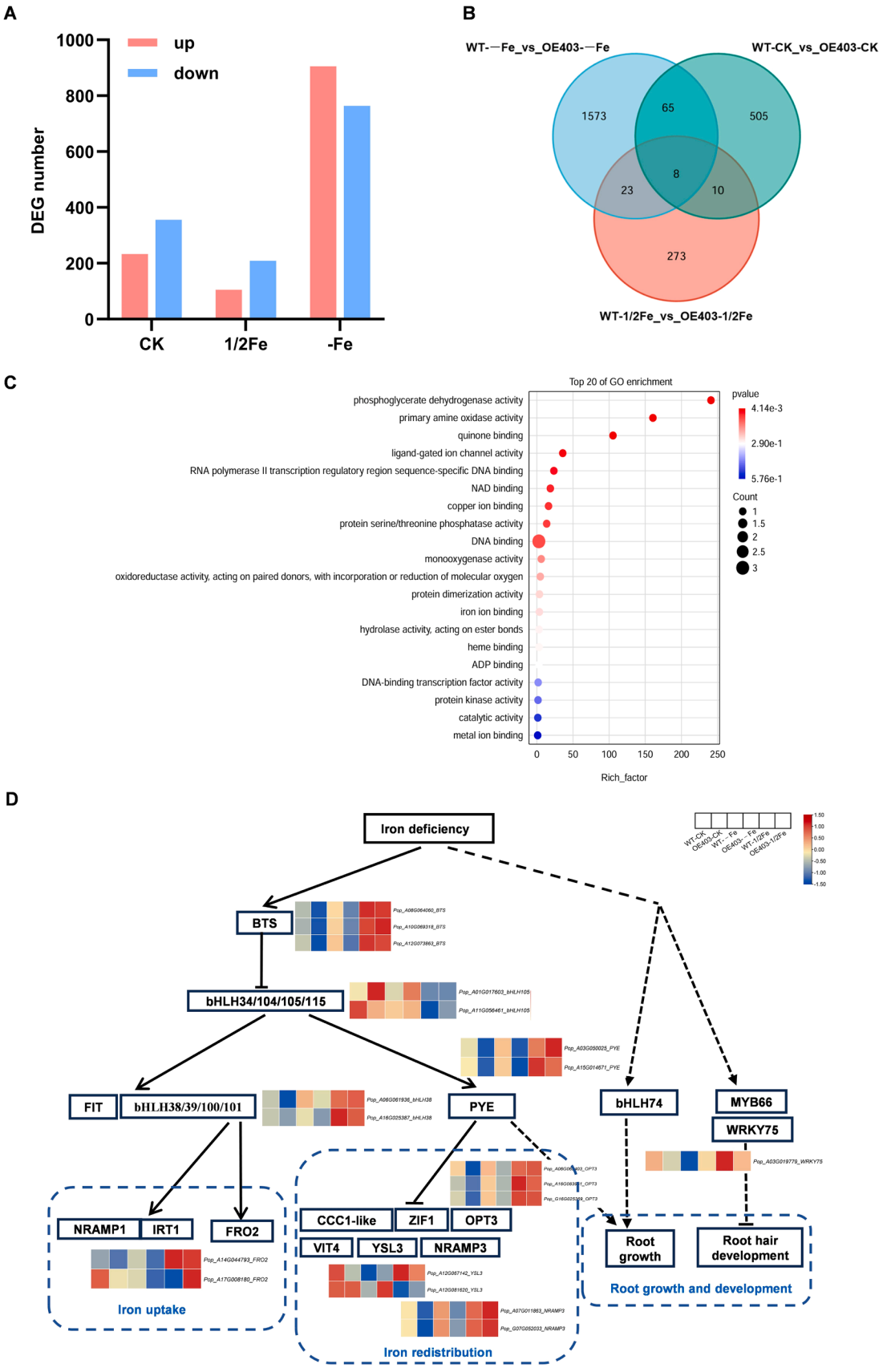


Fig. 7. Overexpression of Nc-miR403 enhances the expression of Fe-deficiency–responsive genes under iron-deficiency stress. (A) Number of differentially expressed genes (DEGs) between OE-miR403 and WT under different Fe concentrations. (B) Venn diagram showing the overlap of DEGs between OE-miR403 and WT under different Fe concentrations. (C) Gene Ontology (GO) enrichment analysis of DEGs in OE-miR403 under iron-deficiency stress. (D) Heatmap of DEGs involved in the iron-deficiency regulatory network.

whole-transcriptome sequencing was performed on root tissues after 12 h of treatment. A total of 263 upregulated and 405 downregulated mRNAs, 54 upregulated and 96 downregulated lncRNAs, and 5 differentially expressed circRNAs (2 upregulated and 3 downregulated) were identified ($|Log_2FC| > 1$, $p < 0.05$). Principal component analysis (PCA) clearly separated CK and FD samples, indicating good reproducibility and distinct transcriptional differences between treatments (Fig. S1).

GO enrichment analysis revealed that differentially expressed genes were mainly associated with catalytic (GO:0003824) and oxidoreductase activities (GO:0016491), highlighting a rapid reconfiguration of cellular redox processes under iron deficiency (Fig. 4A). Notably, genes involved in metal cofactor binding, including iron–sulfur cluster-associated proteins such as *NEET*, were significantly downregulated (Fig. 4B), suggesting impaired cofactor-dependent enzymatic functions. Consistently, KEGG analysis showed significant enrichment in carbon fixation pathways (Fig. S2). Core genes involved in the Calvin cycle (*S17P, ALFP2, KPPR, RBS*) and chlorophyll biosynthesis (*POX2, CHLH*) were coordinately downregulated, together with genes required for PSII stability (*PSBY)* and cyclic electron transport (*PNSB3*) (Fig. 4B and C). These results indicate that iron deficiency rapidly disrupts the photosynthetic processes and carbon assimilation at an early stage.

Interestingly, among 233 identified miRNAs, only three (novel_153, novel_263, and miR403) were upregulated, with miR403 showing the strongest induction, as confirmed by RT–qPCR (Fig. 4D, Fig. S4, Table. S3). Target genes analysis revealed enrichment in hydrolase (GO:0004553) and nucleotidyltransferase activities (GO:0016779), as well as carbohydrate metabolic (GO:0005975) (Fig. S3). KEGG pathways included nicotinate and nicotinamide metabolism, brassinosteroid biosynthesis, and ABC transporters (Fig. S3), suggesting that miRNAs contribute in metabolic reprogramming and stress adaptation under iron deficiency.

Nc-miR403 overexpression enhances iron-deficiency tolerance in poplar

To investigate the role of Nc-miR403, Nc-miR403-overexpressing lines (OE-miR403) were generated in 84K poplar (Fig. 5A and B). As miR403 is evolutionarily conserved to directly target *ARGONAUTE 2* (*AGO2*) transcripts for post-transcriptional silencing, we examined the transcript levels of two poplar *AGO2* homologs (*Pop_A15G084285* and *Pop_G15G010600*) in OE-miR403 lines by RT-qPCR. Both *AGO2* genes were significantly downregulated in OE-miR403#5 and OE-miR403#7 lines compared with WT plants (Fig. 5D).

Under Fe-sufficient conditions (14.68 $mg{\cdot}L^{-1}$, CK), OE-miR403 plants exhibited darker and smaller leaves, with reduced plant height compared with WT. Under moderate Fe-deficiency conditions (7.34 $mg{\cdot}L^{-1}$, 1/2 Fe), plant height differences were reduced, while under severe Fe-deficiency conditions (0 $mg{\cdot}L^{-1}$, −Fe), OE-miR403 plants maintained greener leaves and higher plant height compared to WT, indicating enhanced tolerance to Fe deficiency (Fig. 5C, E). Root architecture was also affected by Fe availability, under Fe-deficiency conditions (−Fe), OE-403 plants exhibited higher total root length, total surface area, average root diameter, and total root volume compared with WT. Importantly, OE-miR403#5 showed a 22.5% increase in average root diameter and 65.3% increase in total root volume (Fig. 5F-I).

Physiological indicators related to oxidative stress were further analyzed. Under Fe deficiency, MDA content was significantly lower in OE-miR403 leaves than in WT, with reductions of 27.27% and 36.58% in OE-miR403#5 and #7, respectively (Fig. 5J). In contrast, antioxidant enzyme activities were enhanced in OE-miR403 plants. POD activity was significantly increased under Fe-deficiency conditions (−Fe) (Fig. 5K). SOD activity was also elevated by 3.80% and 2.00% compared with WT under Fe deficiency (Fig. 5L). Collectively, these results indicate that Nc-miR403 overexpression improves Fe-deficiency tolerance in poplar by enhancing root growth, increasing antioxidant capacity, and reducing oxidative damage.

Nc-miR403 overexpression enhances photosynthetic performance in poplar under iron-deficiency

To investigate the effect of Nc-miR403 overexpression on photosynthetic pigments, the contents of chlorophyll a, chlorophyll b, total chlorophyll, and carotenoids were measured in OE-miR403 and WT plants under different Fe conditions. Under Fe-sufficient conditions (CK), OE-miR403 plants exhibited significantly lower pigment contents than WT. In OE-miR403#5, chlorophyll a, chlorophyll b, total chlorophyll, and carotenoid contents decreased by 18.75%, 26.23%, 20.67%, and 15%, respectively ($p < 0.05$). However, under moderate Fe-deficiency conditions (1/2 Fe), OE-miR403 plants maintained higher chlorophyll and carotenoid levels. Under severe Fe-deficiency conditions (−Fe), both OE-miR403 and WT plants displayed pronounced chlorosis accompanied by a substantial decline in chlorophyll content, with OE-miR403 maintaining a higher chlorophyll level than WT. (Fig. 6B–E).

Gas exchange measurements further revealed that miR403 overexpression altered photosynthetic performance. Under CK conditions, the net photosynthetic rate (Pn), intercellular CO_2 concentration (Ci), stomatal conductance (Gs), and transpiration rate (Tr) of OE-miR403 plants were lower than those of WT (Fig. 6F–I, Table S4). However, under Fe-deficiency conditions (1/2 Fe), all photosynthetic parameters were significantly higher in OE-miR403 plants than in WT. Under Fe-deficiency conditions (−Fe), OE-miR403 plants maintained higher photosynthetic capacity than WT, with Pn increasing by 34.9% and 23.27% in OE-miR403#5 and OE-miR403#7, respectively (Fig. 6F–I). Chlorophyll fluorescence analysis further supported these findings. Under Fe-deficiency conditions (1/2 Fe), OE-miR403 plants exhibited higher Fv/Fm but lower Y(NPQ), NPQ, and qN than WT (Table. S5), indicating reduced photoinhibition and improved maintenance of PSII activity under iron deficiency.

Consistently, chlorophyll fluorescence imaging showed that after one month of treatment under different Fe concentrations, stressed plants developed visible leaf color differences relative to the control group, which became progressively more pronounced with prolonged treatment. These observations were consistent with the gas exchange measurements (Fig. 6A), indicating that Nc-miR403 overexpression improves photosynthetic performance and enhances tolerance to iron-deficiency stress in poplar.

Transcriptome analysis reveals that Nc-miR403 regulates the iron-deficiency response in poplar

To elucidate the regulatory role of Nc-miR403 in plant responses to iron deficiency, RNA-seq analysis was performed on WT and OE-miR403#5 grown under CK, 1/2 Fe, and −Fe conditions. The number of differentially expressed genes (DEGs) between OE-miR403 and WT increased markedly with increasing severity of iron deficiency. Under Fe-deficiency conditions (−Fe), the number of DEGs reached the highest level, with 905 upregulated and 764 downregulated genes (Fig. 7A).

To further characterize the Fe-dependent transcriptional responses, Venn diagram analysis was performed to identify shared and treatment-specific DEGs between OE-miR403 and WT. A total of 8 DEGs were common across all Fe treatments, whereas 23 DEGs were specifically detected under iron-deficient conditions (Fig. 7B). Most DEGs exhibited Fe concentration–dependent expression patterns, while only a small number of core genes were consistently differentially expressed across all treatments. The result suggest that Nc-miR403-mediated transcriptional regulation is strongly dependent on Fe availability.

Functional enrichment analysis of the 23 Fe deficiency–specific DEGs revealed that the majority of enriched Gene Ontology (GO) terms were associated with molecular functions, including DNA binding (GO:0003677), iron ion binding (GO:0005506), and oxidoreductase activity, acting on paired donors, with incorporation or reduction of molecular oxygen (GO:0016705). These results indicate that miR403

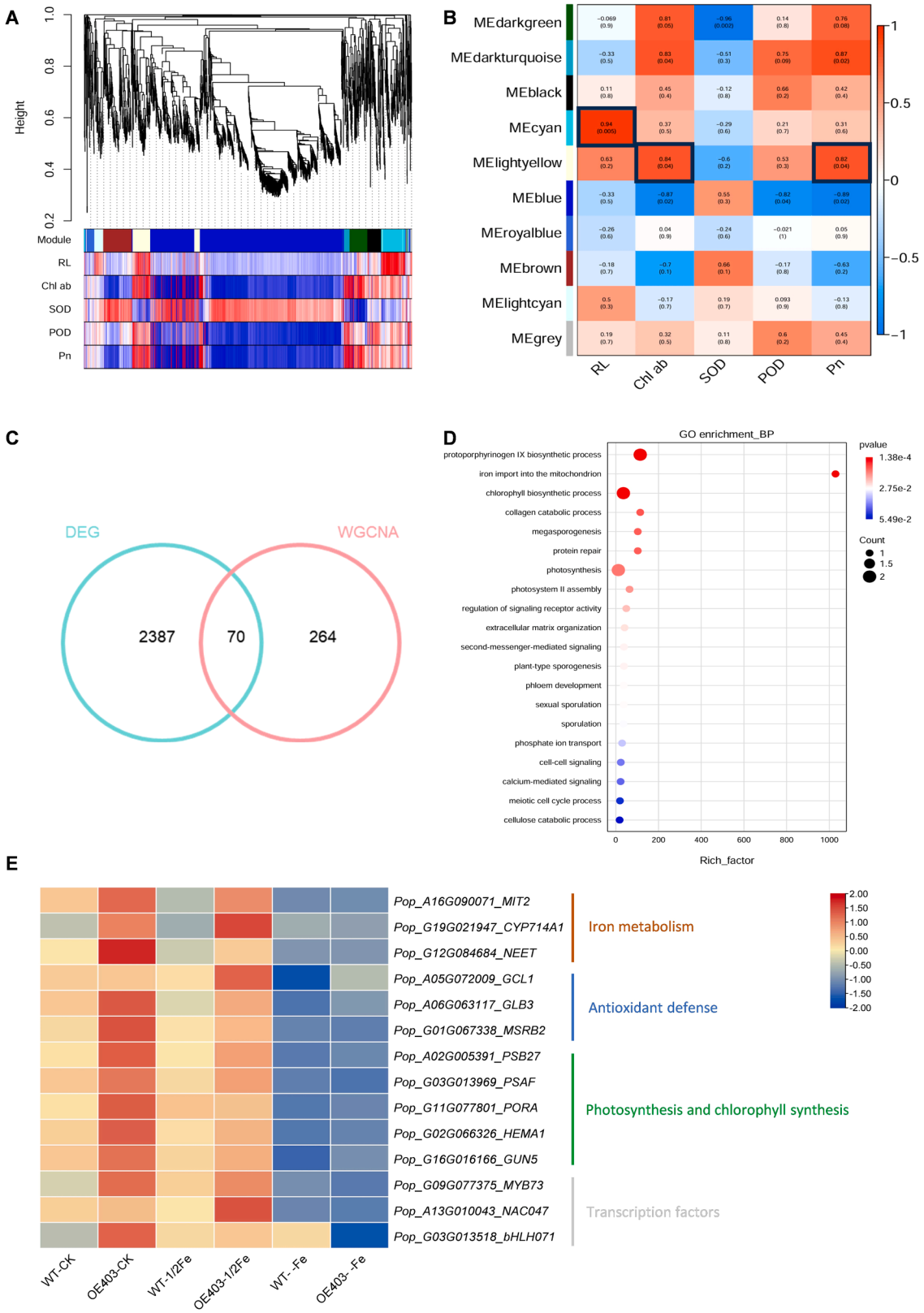


Fig. 8. WGCNA analysis of Nc-miR403-overexpressing poplar under iron-deficiency stress. (A) Hierarchical clustering dendrogram generated by weighted gene co-expression network analysis (WGCNA). (B) Correlation matrix between 10 key WGCNA modules and five traits. (C) Venn diagram showing the overlap between genes in the MEcyan and MElightyellow modules identified by WGCNA and the differentially expressed genes (DEGs) between OE-miR403 and WT under different Fe concentrations. (D) Gene Ontology (GO) enrichment analysis of the overlapping genes between WGCNA modules and DEGs. (E) Heatmap showing the expression patterns of overlapping genes associated with iron-deficiency stress.

may coordinate multiple metabolic and signaling pathways to regulate plant responses to iron deficiency (Fig. 7C).

Poplar roots respond to Fe deficiency in three major processes: Fe uptake, Fe redistribution and root development. Consistent with the enhanced iron-deficiency tolerance observed in OE-miR403 plants, transcriptome analysis revealed that DEGs between OE-miR403 and WT were significantly enriched in the core regulatory network of the iron-deficiency response (Fig. 7D). Transcription factor, *bHLH105* and *PYE* were both upregulated in OE-miR403 plants under iron-deficient conditions. Additionally, genes involved in iron uptake (*IRT1* and *FRO2*), iron redistribution (*NRAMP3* and *YSL3*), and root developmental regulation (*WRKY75*) were significantly upregulated in OE-miR403 plants under Fe deficiency. Collectively, these results suggest that Nc-miR403 overexpression activates key components of the iron-deficiency response network, thereby promoting iron uptake, iron redistribution, and root developmental plasticity, ultimately enhancing plant adaptation to iron-deficiency stress.

WGCNA identifies key gene modules associated with the iron-deficiency response in poplar

To identify key gene modules involved in Nc-miR403 signaling pathway in response to iron-deficiency stress in poplar, transcriptome data from miR403-overexpressing lines and WT plants across different iron concentrations were filtered to remove low-abundance and low-variance genes. A total 2,068 genes were retained for weighted gene co-expression network analysis (WGCNA).

In hierarchical clustering dendrogram, each branch represents a co-expression module, and each leaf corresponds to an individual gene. As shown in Fig. 8A, modules were defined based on the clustering tree, with genes exhibiting high expression similarity grouped within the same module. The heatmap beneath dendrogram represents the correlations between gene modules and five key physiological traits, including root length (RL), chlorophyll content (Chl a+b), superoxide dismutase activity (SOD), peroxidase activity (POD), and net photosynthetic rate (Pn). Darker colors indicate stronger correlations between module genes and the corresponding physiological trait. A total of 10 co-expression modules were identified and assigned distinct colors, including MEdarkgreen, MEdarkturquoise, MEblack, MEcyan, MElightyellow, MEblue, MEroyalblue, MEbrown, MElightcyan, and MEgrey. Correlations analysis revealed that several modules were strongly associated with stress-related physiological traits. Notably, the MEcyan, MEdarkgreen and MElightyellow module exhibited a strong positive correlation with root length (RL) ($r = 0.94$, $p = 0.005$), total chlorophyll content (Chl a+b) ($r = 0.81$, $p = 0.05$) and total chlorophyll content (Chl a+b) ($r = 0.84$, $p = 0.04$), net photosynthetic rate (Pn) ($r = 0.82$, $p = 0.04$), respectively (Fig. 8B).

Among these modules, the MEcyan and MElightyellow modules with relatively higher expression levels were selected for further analysis, comprising a total of 334 genes. These WGCNA module genes were intersected with the DEGs identified between OE-miR403 and WT under varying iron concentrations, resulting in 70 overlapping genes (Fig. 8C). Gene Ontology (GO) enrichment analysis of indicated that these genes were primarily enriched in biological processes, including protoporphyrinogen IX biosynthetic process (GO:0006782), iron import into the mitochondrion (GO:0048250), chlorophyll biosynthetic process (GO:0015995), and photosynthesis (GO:0015979) (Fig. 8D, Table. S6, 7).

Under control (CK) conditions, Nc-miR403 overexpression increased the expression of *MIT2, NEET*, and *MSRB2* genes. As iron concentrations decreased, the expression of these genes declined in both OE-miR403 and WT plants. However, the reduction was less pronounced in OE-miR403 lines. This pattern suggests that Nc-miR403 overexpression confers enhanced tolerance to iron deficiency by maintaining mitochondrial iron homeostasis and antioxidant capacity. Furthermore, photosynthesis-related genes, including *HEMA1, PORA, PSB27*, and *PSAF*, exhibited higher expression levels in OE-miR403 plants compared with WT under CK conditions, and enhanced tolerance to iron deficiency (Fig. 8E).

Discussion

Iron is an essential micronutrient required for plant growth and development, playing critical role in photosynthesis, respiration, and various redox processes (Balk and Schaedler, 2014). However, in most soils iron readily forms insoluble complexes, resulting in low bioavailability for plant uptake (Qing et al., 2015). Consequently, plants frequently experience iron deficiency, which impairs chlorophyll biosynthesis, reduces photosynthetic efficiency, and promotes accumulation of reactive oxygen species (ROS), leading to severe physiological dysfunction. In recent years, increasing attention has been directed towards role of microRNAs (miRNAs) in regulating plant responses to iron-deficiency in different plant species (Kong and Yang, 2010). Nevertheless, our understanding of miRNA-mediated regulatory mechanisms in fast-growing woody species remains limited.

In the present study, seedlings of *N. cadamba* were subjected to short-term iron-deficiency treatment, and responses were analyzed using integrated transcriptome profilingt, including mRNA, miRNA, lncRNA, and circRNA. Three iron-deficiency-responsive miRNAs, miR403, miR-N153, and miR263, were identified in the transcriptome of *N. cadamba*, among which the conserved miR403 was selected for detailed functional characterization. Due to the limited availability of an efficient and stable genetic transformation system in *N. cadamba*, poplar, which possesses a well-established transformation platform and is widely used as a model woody species, was employed as a heterologous expression system. Given that both poplar and *N. cadamba* are woody species, this approach provides a reliable means to elucidate the biological function of Nc-miR403 under iron-deficiency stress. Previous studies have shown that miR403 regulates small RNA–mediated gene silencing by targeting *ARGONAUTE2* (*AGO2*), thereby influencing plant development, and stress responses (Zhang et al., 2023b; Allen et al., 2005). Consistent with this conserved regulatory mode, our results demonstrated that both poplar *AGO2* homologs were significantly downregulated in OE-miR403 lines compared with WT plants, confirming that Nc-miR403 effectively represses *AGO2* transcription in poplar. By generating transgenic poplar plants overexpressing Nc-miR403 and systematically characterizing their phenotypic, physiological, and transcriptomic responses under different iron conditions, this study provides the first evidence that Nc-miR403 overexpression enhances iron deficiency tolerance in poplar, thereby revealing a previously unrecognized role of the conserved Nc-miR403–*AGO2* regulatory module in iron-deficiency adaptation. These findings provide new insights into the adaptive strategies of woody plants and establish the Nc-miR403–*AGO2* module as an important regulator of iron-deficiency responses. Root system plasticity is a key adaptive mechanism enabling plants to cope with nutrient limitations (Pestana et al., 2012). Under iron-deficient conditions, plants often modify root architecture to enhance nutrient acquisition (Ning et al., 2023). Consistent with this, OE-miR403 poplar plants maintained significantly longer roots under iron deficiency, suggesting that Nc-miR403 promotes root growth and enhances the capacity for iron uptake under limiting conditions.

Iron availability strongly influences photosynthetic performance, as it is a critical component of both chlorophyll biosynthesis and the photosynthetic electron transport chain (López-Millán et al., 2016). In this study, OE-miR403 plants maintained higher chlorophyll content and net photosynthetic rate (Pn) under iron-deficient conditions. Chlorophyll fluorescence analysis further revealed higher Fv/Fm values and lower Y(NO) in OE-miR403 plants, indicating reduced photoinhibition and improved photosystem II (PSII) efficiency. Given that iron deficiency disrupts photosynthetic apparatus integrity and reduces photochemical efficiency (Zuo and Zhang, 2009). these findings suggest that Nc-miR403 overexpression contributes to preserving photosynthetic

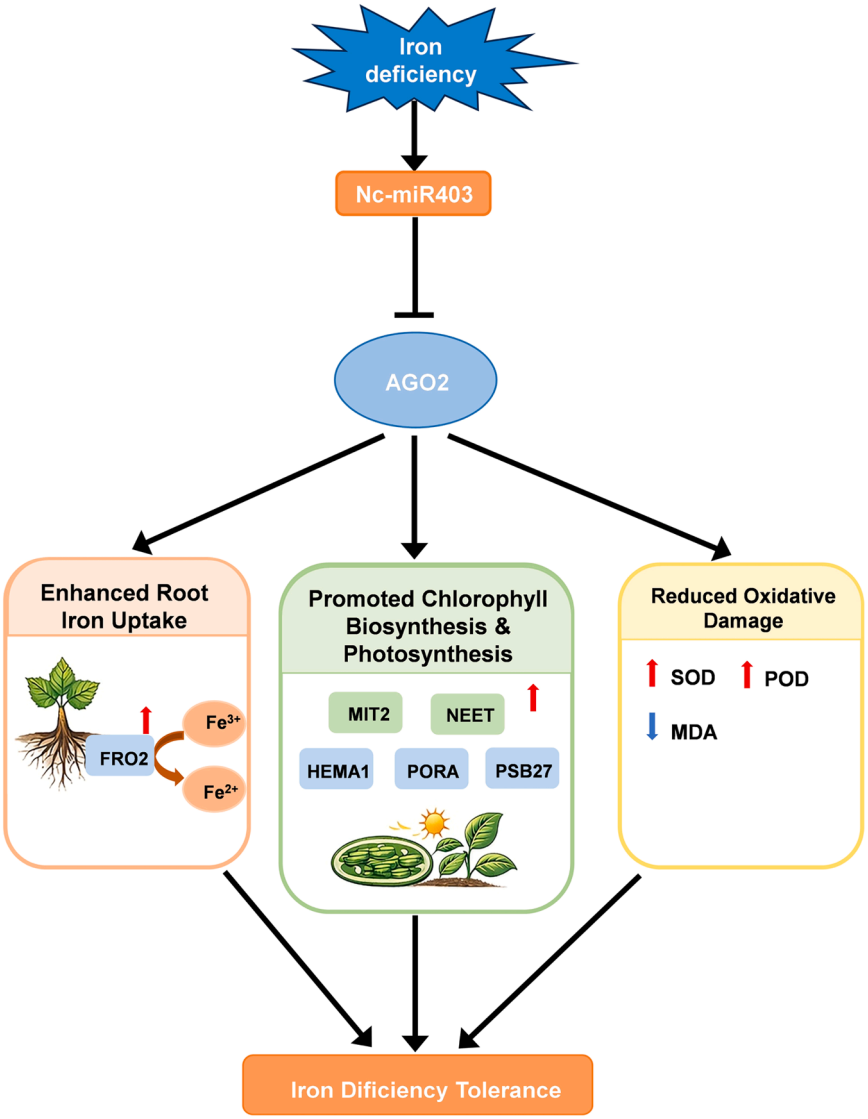


Fig. 9. Proposed regulatory model of miR403 in response to iron deficiency in poplar. Under iron-deficiency stress, Nc-miR403 is induced and directly suppresses *AGO2* expression, thereby relieving *AGO2*-mediated repression on downstream target genes and enabling the activation of stress-responsive pathways. In turn, Nc-miR403 promotes root development, potentially enhancing the ability of plants to acquire iron from the soil. Overexpression of Nc-miR403 significantly upregulates *FRO2*, thereby increasing the reduction of ferric iron (Fe^3) to ferrous iron (Fe^2) and facilitating iron uptake and transport. In addition, Nc-miR403 contributes to the maintenance of mitochondrial iron homeostasis by regulating genes such as *MIT2* and *NEET*. Nc-miR403 also enhances the ROS-scavenging system by increasing antioxidant enzyme activities (SOD and POD) and reducing MDA accumulation, thereby alleviating oxidative damage under iron deficiency. Furthermore, Nc-miR403 promotes the expression of photosynthesis-related genes (*HEMA1, PORA*, and *PSB27*), which helps maintain chlorophyll biosynthesis and photosystem stability. These coordinated responses ultimately enhance the tolerance of poplar to iron-deficiency stress.

stability under stress conditions.

Iron deficiency is also associated with excessive accumulation of ROS, leading to oxidative stress and cellular damage. The antioxidant defense system plays a crucial role in maintaining cellular redox homeostasis (Sandmann, 1985; Gupta et al., 2024). In the present study, OE-miR403 plants exhibited significantly higher activities of SOD and POD compared with WT under iron-deficient conditions., suggesting that Nc-miR403 overexpression enhances the antioxidant capacity of plants. These enzymes are central to ROS scavenging, and their enhanced activity indicates that Nc-miR403 overexpression strengthens antioxidant defenses, thereby improving tolerance to iron-deficiency-induced oxidative stress.

Transcriptomic profiling provides compelling evidence supporting the pivotal regulatory role of Nc-miR403 in to iron deficiency adaptation. The number of DEGs between OE-miR403 and WT increased progressively with the severity of iron limitation, peaking under iron-depleted conditions. While only a small subset of genes was consistently differentially expressed across all treatments, the majority exhibited iron concentration–dependent expression patterns, highlighting the context-dependent nature of Nc-miR403-mediated regulation. Functional enrichment analysis revealed significant overrepresentation of gene ontology terms related to DNA binding, iron ion binding, and oxidoreductase activity, implying that Nc-miR403 orchestrates a broad spectrum of metabolic and regulatory programs under iron stress. Critically, multiple central components of the core iron-deficiency response regulatory network characterized by Chen et al. were significantly upregulated in OE-miR403 plants under iron deficiency (Chen et al., 2019b), including the transcription factors *bHLH105* and *PYE*, as well as genes involved in iron uptake (*IRT1, FRO2*), iron remobilization (*NRAMP3, YSL3*), and root developmental modulation (*WRKY75*). These findings illustrate that Nc-miR403 overexpression reinforces plant adaptation to iron deficiency by activating

the core iron-deficiency signaling cascade, facilitating iron acquisition and redistribution, and augmenting root developmental plasticity.

To further elucidate the molecular mechanisms underlying Nc-miR403-mediated iron-deficiency responses, weighted gene co-expression network analysis (WGCNA) was performed using transcriptome data. A total of 10 co-expression modules were identified, among which the MEcyan and MElightyellow modules showed significant positive correlations with key physiological traits such as root length, chlorophyll content, and net photosynthetic rate. Intersection analysis between module genes and differentially expressed genes identified 70 potential key genes. Gene Ontology (GO) enrichment analysis revealed that these genes were mainly involved in chlorophyll biosynthesis, photosynthesis, and mitochondrial iron transport, suggesting a close relationship between iron metabolism and photosynthetic processes. Several genes associated with iron homeostasis and redox regulation exhibited higher expression in OE-miR403 plants. For example, *MIT2, NEET*, and *MSRB2* showed relatively higher expression in OE-miR403 plants. *MIT2* has been reported to participate in mitochondrial iron transport, whereas *NEET* proteins play important roles in maintaining iron–sulfur cluster homeostasis and cellular redox regulation (Jain et al., 2019; Zandalinas et al., 2022). *MSRB2* functions as a methionine sulfoxide reductase that repairs proteins damaged by ROS (Vieira Dos Santos et al., 2005). Although the expression of these genes decreased under iron-deficiency conditions in both genotypes, the reduction was less pronounced in OE-miR403 plants, suggesting that Nc-miR403 overexpression may enhance plant adaptation to iron deficiency by maintaining mitochondrial iron homeostasis and antioxidant capacity. Furthermore, several photosynthesis-related genes, including *HEMA1, PORA, PSB27*, and *PSAF*, exhibited higher basal expression levels in OE-miR403 plants. *HEMA1* and *PORA* are involved in chlorophyll biosynthesis, whereas *PSB27* and *PSAF* participate in the assembly and stabilization of photosystem II and photosystem I complexes, respectively (Lee et al., 2021; Kumar and Söll, 2000; Chang and Lv, 2024). The relatively higher basal expression of these genes may contribute to maintaining the structural stability of the photosynthetic apparatus under iron-deficiency conditions, thereby ensuring efficient photosynthesis. This finding further elucidates the molecular mechanisms underlying the enhanced photosynthetic performance of OE-miR403 poplar under iron-deficiency conditions.

In summary, this study systematically revealed the important regulatory role of Nc-miR403 in the iron-deficiency response of poplar at phenotypic, physiological, and molecular levels. Overexpression of Nc-miR403 enhances plant tolerance to iron deficiency through multiple mechanisms, including promoting root growth, maintaining photosynthetic efficiency, enhancing antioxidant capacity, and regulating genes involved in iron metabolism and chlorophyll biosynthesis (Fig. 9).

CRediT authorship contribution statement

Minghui Fang: Writing – review & editing, Writing – original draft, Visualization, Validation. **Zhihua Wang:** Investigation, Data curation. **Lu Lu:** Formal analysis, Data curation. **Shangling Jing:** Data curation. **Run Zhang:** Investigation. **Sufang Zhang:** Supervision. **Huiling Li:** Supervision, Funding acquisition. **Qingyin Zeng:** Resources, Funding acquisition, Conceptualization. **Aimin Wu:** Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.stress.2026.101472.

Data availability

The data that has been used is confidential.

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